

Association Between Sad/Hopeless Feelings and Cigarette Use Among Adolescents

1. RESEARCH QUESTION



Is the proportion of current cigarette use different between students who felt sad or hopeless and those who did not?

Motivation: Adolescence is a critical period for the development of health behaviors. This study investigates whether feelings of sadness or hopelessness are associated with a higher prevalence of current cigarette use among adolescents.

2. DATA & VARIABLES

Survey: YRBS_2007

Total Sample Size

13,174
adolescents

Group Variable (Sad_recoded)

- 1 = Felt sad or hopeless (Yes)
- 0 = Did not feel sad or hopeless (No)

Response Variable (Smoke_recoded)

- 1 = Current smoker (codes 2–7)
- 0 = Non-smoker (code 1)

Control / Exploration Variable

- Sex: 0 = Female, 1 = Male

3. METHOD

Statistical Methods:

- Two-Proportion Z-Test
- Linear Probability Model (OLS Regression)

Hypotheses (Two-Proportion Z-Test / LPM Slope)

$H_0 : \beta_1 = 0$ (No association)

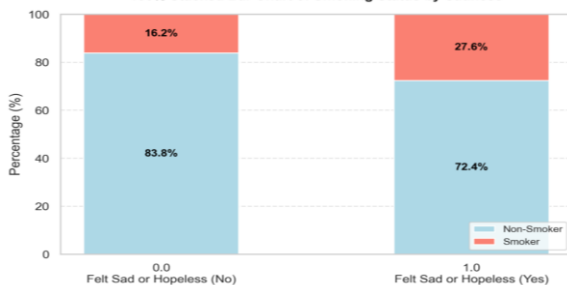
$H_a : \beta_1 \neq 0$ (Association exists)

Significance Level: $\alpha = 0.05$



4. 100% STACKED BAR CHART OF SMOKING STATUS BY SADNESS

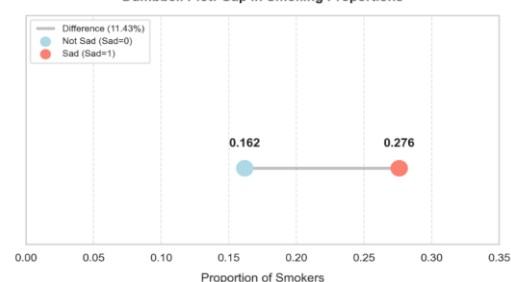
100% Stacked Bar Chart of Smoking Status by Sadness



Interpretation: 27.6% of students who felt sad or hopeless are current smokers compared to 16.2% among those who did not.

5. DUMBBELL PLOT: GAP IN SMOKING PROPORTIONS

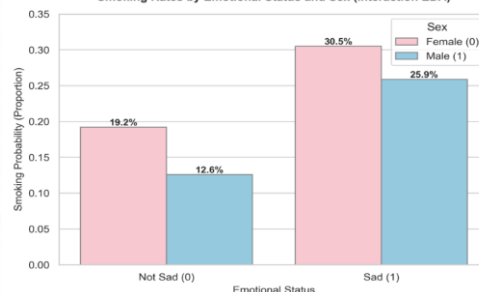
Dumbbell Plot: Gap in Smoking Proportions



Interpretation: The proportion of smokers among students who felt sad or hopeless is **11.43** percentage points higher.

6. SMOKING RATES BY EMOTIONAL STATUS AND SEX (INTERACTION EDA)

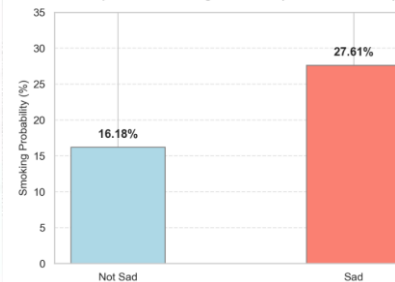
Smoking Rates by Emotional Status and Sex (Interaction EDA)



Interpretation: Sadness is associated with higher smoking rates for both females and males. The effect is consistent across both sexes.

7. COMPARISON OF SMOKING PREVALENCE (RELATIVE RISK BASIS)

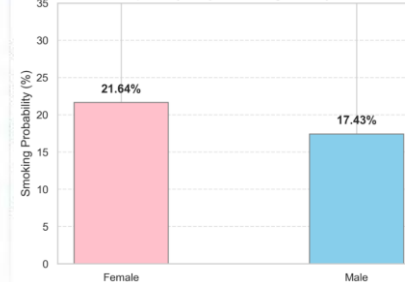
Comparison of Smoking Prevalence (Relative Risk Basis)



Interpretation: Students who felt sad or hopeless are 1.71 times more likely to be current smokers than those who did not.

8. EXPLORATORY ANALYSIS: SMOKING RATES BY SEX

Exploratory Analysis: Smoking Rates by Sex



Interpretation: Overall, females (21.64%) have higher smoking rate than males (17.43%) in this sample.

9. 95% CONFIDENCE INTERVAL FOR THE REGRESSION COEFFICIENT (SLOPE)

--- No Effect (Slope = 0)
— Sad_recoded Effect (Slope)



Interpretation: The 95% confidence interval for the slope is [0.100, 0.129]. Because the entire interval is above zero and does not cross the No Effect line, the positive association is statistically significant and highly robust.

10. CONCLUSION



Adolescents who reported feeling sad or hopeless had a smoking prevalence of 27.61%, compared with 16.18% among those who did not. The prevalence was **11.43 percentage points** higher in the sad/hopeless group (95% CI: 0.100–0.129, $p < 0.001$). These findings indicate a strong statistical association between emotional distress and current cigarette use.



IMPORTANT CODE (PYTHON)

```
import statsmodels.formula.api as smf
df_clean = df[['Smoke_recoded', 'Sad_recoded']].dropna()
model = smf.ols('Smoke_recoded ~ Sad_recoded', data=df_clean).fit()
print(model.summary()) #  $\beta_1 = 0.1143$ , p-value = 3.27e-51
```

KEY MODEL RESULTS (MAIN MODEL)

Intercept (β_0)	Slope (β_1)	p-value	Relative Risk (RR)	Difference in Proportions
0.1618 (16.18%)	0.1143 (11.43%)	3.27e-51 (< 0.001)	1.71x	11.43 percentage points
95% CI for β_1 : [0.100, 0.129]				

DATASET INFORMATION

- Source: Youth Risk Behavior Survey
- Sample: 13,174 adolescents
- Variables: SadOrHopeless, CurrentCigaretteUse, Sex
- Method: Two-Proportion Z-Test & Linear Probability Model (OLS)

